

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.02, beta.var.scale = 1\n')
```

```
## sdev = 0.02, beta.var.scale = 1
```

Key Summary Statistics

```
cat(paste0('```\n', '

--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.4756  0.5028  0.5100  0.5104  0.5178  0.5500

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.5757  0.7111  0.6569  0.7989  0.9650
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0000  0.5617  0.6956  0.6403  0.7806  0.9311
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.01667 0.04500 0.07111 0.23756 0.26667 1.00000
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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02056 0.04556 0.06889 0.20820 0.20542 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01722 0.05278 0.13500 0.33918 0.63889 1.00000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.2039  0.2228  0.2311  0.2310  0.2383  0.2661

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.3917  0.5628  0.5089  0.6772  0.9361
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.3822  0.5478  0.4929  0.6556  0.8678
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01833 0.44500 0.50471 1.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.008333 0.047222 0.331375 0.869028 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.6094  1.0000  0.7691  1.0000  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.1133  0.1361  0.1433  0.1435  0.1500  0.2106

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beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1735  0.3856  0.3567  0.5306  0.9006
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1772  0.3850  0.3483  0.5188  0.8261
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.9344  1.0000  0.8281  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.000000 0.007222 0.278952 0.666944 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.4706  1.0000  0.7516  1.0000  1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.03333 0.05167 0.05833 0.05966 0.06611 0.12500

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1672  0.3858  0.3498  0.5217  0.8906
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1722	0.3900	0.3454	0.5178	0.7950

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.7589	0.9589	0.7873	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0050	0.3091	0.8474	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	1.000	1.000	0.894	1.000	1.000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.02833	0.04611	0.05333	0.05527	0.06056	0.13778

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06278	0.29056	0.27741	0.43778	0.85944

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06667	0.29611	0.27435	0.43889	0.74167

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4794	0.8161	0.6891	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8694	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
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    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2861  0.7485  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.8872  1.0000  0.8592  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.02167 0.03889 0.04667 0.04834 0.05500 0.13111

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.04667 0.26222 0.25582 0.40833 0.83333
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1      1      1      1      1      1

gamma.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.05111 0.27000 0.25169 0.40611 0.74278
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.4256  0.4716  0.9322  1.0000
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.7356 1.0000  1.0000  0.9997  1.0000  1.0000
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2331  0.4537  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.7939  1.0000  0.8188  1.0000  1.0000

--- Summary for K = 7 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.02611 0.04444 0.05333 0.05493 0.06167 0.13000

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

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0.0000 0.0100 0.2078 0.2192 0.3683 0.7933
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1         1      1         1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1         1      1         1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1         1      1         1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1         1      1         1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.01111 0.21333 0.21385 0.36611 0.67389
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.05833 0.33198 0.72611 1.00000
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.4089  1.0000  1.0000  0.9985  1.0000  1.0000
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.1776  0.1468  1.0000
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000  0.5406  0.8644  0.7100  1.0000  1.0000

--- Summary for K = 8 ---
Z.ar:
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.04889 0.07833 0.09333 0.09650 0.10944 0.18444

beta.ar:
Modality 1 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
0.0000  0.0050  0.1183  0.1609  0.2806  0.7589
Modality 2 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1         1      1         1      1      1
Modality 3 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1         1      1         1      1      1
Modality 4 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1         1      1         1      1      1
Modality 5 :
  Min. 1st Qu. Median      Mean 3rd Qu.  Max.
    1         1      1         1      1      1

```

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gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.005556 0.126111 0.160385 0.286667 0.632778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.1552 0.2444 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.9411 1.0000 0.9596 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.06737 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.0000 0.3617 0.3706 0.6639 1.0000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.2750 0.3117 0.3217 0.3245 0.3333 0.4433

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.01111 0.09111 0.17299 0.24944 0.80000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.01222 0.07889 0.11276 0.18444 0.61389
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.10427 0.06056 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000 0.3417 0.4600 0.4531 0.5728 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.01746 0.00000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.

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0.0000 0.0000 0.0000 0.1399 0.1744 1.0000
', '\n`'))
```

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## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4756 0.5028 0.5100 0.5104 0.5178 0.5500
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5757 0.7111 0.6569 0.7989 0.9650
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5617 0.6956 0.6403 0.7806 0.9311
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01667 0.04500 0.07111 0.23756 0.26667 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02056 0.04556 0.06889 0.20820 0.20542 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01722 0.05278 0.13500 0.33918 0.63889 1.00000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2039 0.2228 0.2311 0.2310 0.2383 0.2661
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.0000 0.3917 0.5628 0.5089 0.6772 0.9361
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.3822 0.5478 0.4929 0.6556 0.8678
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.01833 0.44500 0.50471 1.00000 1.00000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.008333 0.047222 0.331375 0.869028 1.000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.6094 1.0000 0.7691 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.1133 0.1361 0.1433 0.1435 0.1500 0.2106
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.1735 0.3856 0.3567 0.5306 0.9006
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1772 0.3850 0.3483 0.5188 0.8261
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.9344 1.0000 0.8281 1.0000 1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.007222 0.278952 0.666944 1.000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.4706 1.0000 0.7516 1.0000 1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.03333 0.05167 0.05833 0.05966 0.06611 0.12500
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1672 0.3858 0.3498 0.5217 0.8906
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1722 0.3900 0.3454 0.5178 0.7950
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.7589 0.9589 0.7873 1.0000 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0050 0.3091 0.8474 1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000 1.000 1.000 0.894 1.000 1.000

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##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02833 0.04611 0.05333 0.05527 0.06056 0.13778
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.06278 0.29056 0.27741 0.43778 0.85944
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.06667 0.29611 0.27435 0.43889 0.74167
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.4794  0.8161  0.6891  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.8694  1.0000  1.0000  1.0000  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2861  0.7485  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.8872  1.0000  0.8592  1.0000  1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02167 0.03889 0.04667 0.04834 0.05500 0.13111
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.04667 0.26222 0.25582 0.40833 0.83333
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.05111 0.27000 0.25169 0.40611 0.74278
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.4256  0.4716  0.9322  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.7356  1.0000  1.0000  0.9997  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2331  0.4537  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.7939  1.0000  0.8188  1.0000  1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02611 0.04444 0.05333 0.05493 0.06167 0.13000
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0100  0.2078  0.2192  0.3683  0.7933
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01111 0.21333 0.21385 0.36611 0.67389
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.05833 0.33198 0.72611 1.00000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4089  1.0000  1.0000  0.9985  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1776  0.1468  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5406  0.8644  0.7100  1.0000  1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.04889 0.07833 0.09333 0.09650 0.10944 0.18444
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0050  0.1183  0.1609  0.2806  0.7589
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.126111 0.160385 0.286667 0.632778
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1552  0.2444  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.9411  1.0000  0.9596  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.06737 0.00000 1.00000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.3617  0.3706  0.6639  1.0000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2750  0.3117  0.3217  0.3245  0.3333  0.4433
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01111 0.09111 0.17299 0.24944 0.80000
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01222 0.07889 0.11276 0.18444 0.61389
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.10427 0.06056 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.3417 0.4600 0.4531 0.5728 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.01746 0.00000 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.1399 0.1744 1.0000
##
## ‘ ‘ ‘

```

Conclusion

and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.